Triangle Congruence: Solving for Corresponding Parts Answer Key

High School Geometry

Quick Answers

1. 5	3. 8	5. 5, 22	7. 8
2. 15	4. 67	6. 6, 25	8. 13

Curriculum

Level: High School Geometry
HSG-CO.A–HSG-CO.D – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 12 tagged checks.

1. $\triangle ABC \cong \triangle DEF$, $AB = 3x + 2$, $DE = 17$. Reading the statement in order, $A \leftrightarrow D$ and $B \leftrightarrow E$, so $\overline{AB}$ corresponds to $\overline{DE}$ and CPCTC gives $3x+2 = 17$. Subtract 2: $3x = 15$. Divide by 3: $\boxed{x = 5}$

2. $\triangle PQR \cong \triangle STU$, $m\angle P = 4y - 10$, $m\angle S = 50$. Since $P \leftrightarrow S$, the two angle measures are equal: $4y - 10 = 50$. Add 10: $4y = 60$. Divide by 4: $\boxed{y = 15}$

3. $\triangle JKL \cong \triangle MNP$, $JK = 2x + 7$, $MN = 4x - 9$. Here $J \leftrightarrow M$ and $K \leftrightarrow N$, so $\overline{JK}$ corresponds to $\overline{MN}$: $2x + 7 = 4x - 9$. Subtract $2x$: $7 = 2x - 9$. Add 9: $16 = 2x$. Divide by 2: $\boxed{x = 8}$
(Check: both sides measure $2(8) + 7 = 23$ inches.)

4. $\triangle ABC \cong \triangle DEF$ with $m\angle A = 52$ and $m\angle B = 61$. The third letters correspond, $C \leftrightarrow F$, so $m\angle F = m\angle C$. The angle sum of $\triangle ABC$ gives $52 + 61 + m\angle C = 180$, so $113 + m\angle C = 180$ and $m\angle C = 67$. By CPCTC that is also the measure of $\angle F$: $\boxed{m\angle F = 67^\circ}$

5. $\triangle RST \cong \triangle XYZ$, $m\angle R = 5x - 3$, $m\angle X = 2x + 12$.
(a) $R \leftrightarrow X$, so $5x - 3 = 2x + 12$. Subtract $2x$: $3x - 3 = 12$. Add 3: $3x = 15$. Divide by 3: $\boxed{x = 5}$
(b) Substitute back into either expression: $5(5) - 3 = 22$, and the check $2(5) + 12 = 22$ agrees. $\boxed{m\angle R = 22 \text{ degrees}}$

6. **Bridge truss.** $\triangle ABC \cong \triangle DEF$, $AC = 4x + 1$ and $DF = 2x + 13$ metres.
(a) $A \leftrightarrow D$ and $C \leftrightarrow F$, so $\overline{AC}$ corresponds to $\overline{DF}$: $4x + 1 = 2x + 13$. Subtract $2x$: $2x + 1 = 13$. Subtract 1: $2x = 12$. Divide by 2: $\boxed{x = 6}$
(b) $AC = 4(6) + 1 = 25$, and the check $DF = 2(6) + 13 = 25$ agrees. $\boxed{AC = 25 \text{ m}}$

7. (a) **Model two-column proof (open reasoning — review by hand).**

Statements	**Reasons**
1. $\overline{AB} \cong \overline{CD}$	1. Given
2. $\angle BAC \cong \angle DCA$	2. Given
3. $\overline{AC} \cong \overline{CA}$	3. Reflexive property of congruence
4. $\triangle BAC \cong \triangle DCA$	4. SAS (sides in statements 1 and 3 with the included angle in statement 2)

The included angle matters: $\angle BAC$ sits between $\overline{AB}$ and $\overline{AC}$, and $\angle DCA$ sits between $\overline{CD}$ and $\overline{CA}$, so the pairs are matched side–angle–side and not side–side–angle.

(b) The congruence $\triangle BAC \cong \triangle DCA$ matches $B \leftrightarrow D$, $A \leftrightarrow C$, $C \leftrightarrow A$, so $\overline{BC}$ corresponds to $\overline{DA}$ and CPCTC gives $3x - 4 = 20$. Add 4: $3x = 24$. Divide by 3: $\boxed{x = 8}$

8. Find and fix the mistake.
(a) **Model explanation (open response — review by hand).** The statement $\triangle ABC \cong \triangle DFE$ matches letters by position: $A \leftrightarrow D$, $B \leftrightarrow F$, $C \leftrightarrow E$. Kim used the alphabetical order of the second triangle's name instead of its written order, which is what produced $\overline{DE}$. Since $A \leftrightarrow D$ and $B \leftrightarrow F$, the side that corresponds to $\overline{AB}$ is $\overline{DF}$. (The side $\overline{DE}$ corresponds to $\overline{AC}$, because $C \leftrightarrow E$.)
(b) The correct equation is therefore $AB = DF$, that is $2x + 5 = 31$. Subtract 5: $2x = 26$. Divide by 2: $\boxed{x = 13}$
(Check: $2(13) + 5 = 31$ centimetres, matching DF.)